Tristen Chen

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EDUCATION & AWARDS

University of Toronto - *B.A.Sc. in Mechanical Engineering + PEY Co-op* Sept 2024-June 2029

- **Awards:** Dean's Honour List, Wallace Chalmers Engineering Design Award, 1st in Battle of the School Hackathon simulation track
- **Certificates:** SolidWorks CSWA, Basic Machining Certificate
- **Relevant Coursework:** Fundamentals of Computing, Mechanical Engineering Design, Material Science, Thermodynamics, Fluids, Kinematics of Machines, Market & Competitive Strategy, Design Optimization, Heat & Mass Transfer

SKILLS

- **CAD:** Fusion 360, Inventor, AutoCAD, **SolidWorks (CSWA)**, GD&T
- **Mechanical Design:** BOM Generation, **Failure Analysis**, Gearbox Design, Finite Element Analysis, **Component Sourcing**
- **Mechatronics:** ESP32, UART, **Servo/Stepper Motors Control**, Circuit Prototyping
- **Manufacturing & Prototyping**: Machining (Turning/ Milling), **FDM 3D printing**, Soldering
- **Programming & Software:** C/C++, MATLAB, **Python (OpenCV, NumPy, SciPy),** Notion, Slack, Framer, CapCut
- **Professional Skills:** Microsoft Office, Technical Report Writing, Communication, Team Coordination

EXPERIENCE

Senior Optomechanical Engineer - *Payload Subteam, UTAT Space System, University of Toronto,* Jan 2026–Present

- Engineered a venting system with a 3x safety factor for internal pressure of the optical lens housing, preventing lens fracture due to launch pressure differential. Researched NASA and NANOCUB standards for hole-to-volume ratio, evaluated manufacturing methods, and produced technical reports documenting the process.
- Directed SOLIDWORKS version control for a 20-person engineering team using Sibe PDM's check-out system, repairing broken mates, and ensuring all subsystem modifications were integrated without redundancy.

Team Lead - *EWB Smart Watering System, University of Toronto* Oct 2025-April 2026

- Led a 4-person cross-disciplinary team through the full product lifecycle to automate the watering process for the Hart House Community Garden. Prepared weekly meetings, created a Gantt chart with project milestones, and delivered a functional autonomous prototype and slideshow for the end-of-year conference.
- Self-taught and implemented C/C++ control loops for an ESP32 data acquisition system, logging and processing moisture and temperature readings for dynamic pump actuation.
- Coordinated stakeholder communication through Slack. Translated community garden needs into engineering requirements and produced bi-weekly reports of timelines and dependencies to the Technical Branch.

Automotive Manufacturing Intern - *Taiwan Isuzu Motors Co., Ltd* May 2025-Aug 2025

- Participated in a 20+ person production team semi-knocked-down (SKD) manufacturing workflow, facilitating the final assembly of 10+ commercial trucks daily.
- Conducted rigorous quality control inspections on the active production line, interpreting complex assembly drawings to prevent defects and ensure strict compliance with Isuzu manufacturing standards.

Fabrication Team Member – *Chem-E-Cube, University of Toronto* Dec 2024–Oct 2025

- Independently took ownership of the cooling architecture for a Direct Air Capture Cube, optimizing the Monolith's CO2 absorption rate to secure a top 15 finish (13th/60+ teams) and qualify for the international finals in Boston.
- Researched and developed a current-control strategy for sourced Peltier plates, integrating them with custom 3D-printed air channels to successfully achieve 160% cooling efficiency.
- Designed airtight CO2 and temperature sensor housings in Inventor to facilitate reliable data acquisition during operation. Utilized design for additive manufacturing principles to eliminate print supports and integrated locator pins for epoxy joining.

Team Captain - *VEX Robotics (96969Y), Pacific American School, Taiwan* Sept 2022-Aug 2024

- Directed a 7-person team (Mechanical, Software, Marketing, and Drive team) to become one of only 5 Taiwan Teams to qualify for the VEX Championship, capturing a top-10 global skills score (Spin Up) and winning the Design Award at Taiwan's only international competition (Over Under).
- Engineered a pioneering 3-level climbing robot (High Stakes) by benchmarking existing FRC designs, modeling the mechanism in Fusion 360, and fine-tuning the physical prototype to adapt to the perfectly vertical triangular poles using pistons, winch, and power take-off (PTO) mechanisms.
- Programmed autonomous routes using odometry and custom PID control loops for high-precision chassis movement. Implemented Bang-Bang control for rapid flywheel recovery, allowing for Triple-shoots in under 0.5 seconds. (Spin Up).

PROJECTS

Project Lead - *Stroke Bot (Battle of the School 24 hr Hackathon 1st in Simulation Track)* Sept 2026-Present

- Co-led a 2-person team to architect a vision-to-drawing simulation for Bracket Bot, winning 1st place out of 15 teams at a 24-hr hackathon. Conceptualized and expanded the project into an ongoing physical build of an ultra-fast CNC Drawing machine.
- Translated OpenCV-extracted image contours into individual joint poses, utilizing Python libraries SciPy and NumPy to create an inverse kinematics solver that incorporates warm starting to address kinematic redundancies.
- Validated robotic kinematics using Rerun for 3D visualization, establishing target operational speeds to source high-performance servos and generate a BOM for the physical extension.

Project Founder - *Soft Robotics Telepresence System* April 2026-Present

- Identified a gap in remote physical interaction and founded an independent project to engineer a tendon-driven telepresence plush, successfully showcasing the prototype at a Socratica session.
- Directed a 3-person multidisciplinary team through 4 rapid hardware iterations, establishing cross-functional development timelines and workflow between the mechanical, electrical, and software subteams.
- Designed and 3D-printed a compact 6:1 planetary gearbox for NEMA 17 Steppers to achieve a 1.8Nm holding torque through Fusion 360 Gear Generator and SolidWorks, getting 150+ downloads on GrabCAD.
- Diagnosed lateral-torsional buckling in an initial 3D-printed leaf spring through physical testing, resolving the failure by redesigning the compliant joint with an anisotropic FR4 fiberglass spine. Validated the new design against stall torque and fatigue failure using SolidWorks FEA and first-principles-based hands-on calculation.
- Reduced the system's horizontal footprint by 30% and minimized tendon path-length variation by optimizing the spatial layout, designing a double-layer motor stack and Bowden cable routing to pass directly through the shoulder joint's center of rotation.
- Engineered a real-time computer vision pipeline in Python (MediaPipe), reducing end-to-end latency to sub-300ms by multithreading camera capture, migrating data transmission to a lightweight UDP protocol, and fine-tuning a Kalman filter to smooth noisy joint coordinate data.
- Authored technical documentation to justify engineering milestones and material selections, alongside developing a product landing page (Framer) and promotional video (CapCut) to effectively communicate the system's value proposition to non-technical stakeholders.

Individual Coursework - *Numerical Methods Heat Transfer Solver (MATLAB)* Feb 2026-March 2026

- Engineered custom numerical solvers in MATLAB to simulate complex physical phenomena, evaluating steady-state convective cooling efficiency and modeling fluid droplet surface mechanics.
- Discretized a 1D thermal model for an aluminum heatsink fin using finite volume formulation, programming the Thomas Algorithm (TDMA) and a Secant root-finding method to calculate the exact minimum convective heat transfer coefficient required to prevent component melting.
- Programmed a least-squares polynomial regression algorithm to model droplet boundary profiles (evaluating 2nd through 9th order fits), optimizing R^2 values while preventing endpoint distortion to accurately estimate surface hydrophobicity.
- Computed the Eötvös number to evaluate surface tension versus gravity, and implemented multi-segment Trapezoidal and Simpson's 1/3 rules to numerically integrate fitted polynomials for accurate total droplet volume estimation.

Project Co-Lead - *MIE243 Cinematography Robotic Arm Design Project* Sept 2025-Dec 2025

- Co-led the mechanical design of a 5-DOF cinematography robotic arm, developing project schedules and managing a comprehensive BOM in Microsoft Excel to ensure strict development deadlines were met.
- Defined kinematic requirements, physical load cases, and safety features for a 5-DOF arm through benchmarking against cinematography and industrial robotic systems and applying relevant ISO standards.
- Executed the full-cycle mechanical design of a motorized mounting interface, delivering a 20+ part CAD assembly and complete GD&T manufacturing drawings for final production.